ECE 330 · Fall 2026

HW7
完整题目截图与解答

共 7 题 · 题面截图、中文题意、作答图与逐步推导

> 按现有知非录的实例参数整理。题面保留实际网页截图；文字解答为既有独立解析，非官方解答。截图中的输入值保留网页原貌，以后续解答中的结果为准。

1. HW7.1 二次磁链：非线性能量和正向力
2. HW7.2 分两段：电端回送能量与机械端零做功
3. HW7.3 闭合循环：先用恒流段连接状态
4. HW7.4 先固定磁链求力，再沿恒流路径积分
5. HW7.5 λ−i折线：有符号梯形积分
6. HW7.6 二次电流本构：两端状态的磁能
7. HW7.7 非线性折线：由能量守恒求机械输入

整理日期 2026-11-04
来源：ECE330 知非录 HW7 现行阅读版与配套题面截图。

HW7.1 二次磁链：非线性能量和正向力

完整原题截图

HW7.1. SP 7.1 (part 1)

The flux linkage in an energy-conservative electromechanical system is given by

$$\lambda(i,x) = \frac{1}{2}L(x)i^2$$

where $L(x) = 2x + 3$ H/A. Here, x is the size of the gap in m, and i is the current applied to the coil in A.

Compute the energy (in Joules) in the system at $i = 2$ A and $x = 14$ m.

Wm [J]= number ±0.5%

Compute the force of electrical origin (in Newtons) at $i = 2$ A and $x = 14$ m.

f [N]= number ±0.5%

来源：PrairieLearn HW7.1 实际网页题面。2026-11-02 截图。

HW7.1 二次磁链：非线性能量和正向力 · 解答

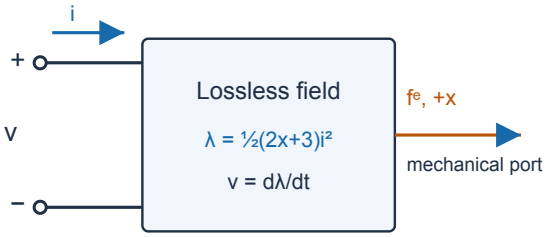
完整中文题意

保守机电系统满足λ(i,x)=½L(x)i²，L(x)=2x+3，单位H/A；x是气隙尺寸，单位m，i是线圈电流，单位A。i=2 A、x=14 m时，求(a)系统储能Wm，单位J；(b)电磁力f，单位N。保留原题的14 m，不改成cm。

作答图与逐步求解

本题作答要画下列模型图，并保留图中方向、坐标、极性和参数。 原题截图不能代替作答图。

HW7.1 Quadratic flux linkage: keep the nonlinear model



L(x) has units H/A, not H.

W′ = ∫₀ⁱ λ(î,x)dî = L(x)i³/6

W = iλ − W′ = L(x)i³/3

fe = (∂W′/∂x)i = i³/3

i = 2 A; x = 14 m; L = 31 H/A

fe > 0: along increasing x →

Electrical → field: EFE = ∫ i dλ

Mechanical → field: EFM = −∫ fe dx

ΔWm = EFE + EFM

$$L(14) = 2(14) + 3 = 31 \text{ H/A},$$

$$W'_m = \int_0^i \frac{1}{2}L(x)\hat{i}^2\, d\hat{i} = \frac{L(x)}{2}\left[\frac{\hat{i}^3}{3}\right]_0^i = \frac{L(x)i^3}{6},$$

$$W_m = i\lambda - W'_m = \frac{1}{2}L(x)i^3 - \frac{1}{6}L(x)i^3 = \frac{1}{3}L(x)i^3$$

$$= \frac{[2(14)+3](2)^3}{3} \approx 82.6667 \text{ J}.$$

$$f^e = \left(\frac{\partial W'_m}{\partial x}\right)_i = \frac{i^3}{6}\frac{d(2x+3)}{dx} = \frac{i^3}{3} = \frac{(2)^3}{3} \approx +2.66667 \text{ N}.$$

正号沿+x，使题定义的间隙尺寸增大。核验：λ=½(31)(2²)=62 Wb·turn；W′=41.333… J，W+W′=124 J=iλ。

HW7.2 分两段：电端回送能量与机械端零做功

完整原题截图

HW7.2. SP 7.1 (part 2)

The flux linkage in an energy-conservative electromechanical system is given by

$$\lambda(i,x) = \frac{1}{2}L(x)i^2$$

where $L(x) = 2x + 3$ H/A. Here, x is the size of the gap in m, and i is the current applied to the coil in A.

If the gap changes from $x = 14$ m to $x = 5$ m while maintaining a fixed current $i = 11$ A, what is the energy transfered to the magnetic field from the electrical system (EFE) for this trajectory?

EFE [J]= number ±0.5%

If the current is then changed from $i = 11$ A to $i = 16$ A while maintaining a fixed gap $x = 5$ m, what is the energy transfered to the magnetic field from the mechanical system (EFM) for this trajectory?

EFM [J]= number ±0.5%

来源：PrairieLearn HW7.2 实际网页题面。2026-11-02 截图。

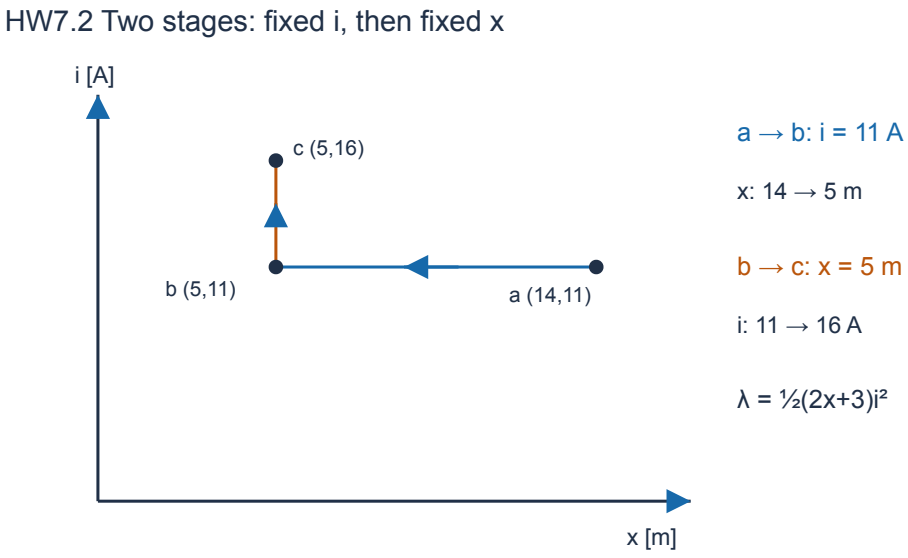
HW7.2 分两段：电端回送能量与机械端零做功 · 解答

完整中文题意

仍有λ=½L(x)i²、L(x)=2x+3 H/A。第一段保持i=11 A，令x从14 m变到5 m，求从电系统传入磁场的能量EFE；第二段接着保持x=5 m，令i从11 A变到16 A，求从机械系统传入磁场的能量EFM。两问是不同的两段轨迹，单位J。

作答图与逐步求解

本题作答要画下列模型图，并保留图中方向、坐标、极性和参数。 原题截图不能代替作答图。



Stage 2: dx = 0 ⇒ EFM = 0, even though current and field energy change.

(a) 第一段i=11 A，端点的L从31变为13 H/A。

$$\mathrm{EFE}_{a\to b}=\int_{\lambda_a}^{\lambda_b} i\,d\lambda = i(\lambda_b-\lambda_a)=\frac{i^3}{2}[L(5)-L(14)]$$
$$=\frac{(11)^3}{2}\{[2(5)+3]-[2(14)+3]\}=\frac{1331}{2}(13-31)=-11979\ \mathrm{J}.$$

负号表示磁场向电系统回送11979 J；不能报正的传入量。

(b) 第二段保持x=5 m，因dx=0，

$$\mathrm{EFM}_{b\to c}=-\int_b^c f^e\,dx=-\int_b^c f^e(0)=0\ \mathrm{J}.$$

检查第一段：$\Delta W_m = i^3\Delta L/3 = -7986$ J；机械输入+3993 J；−11979+3993=−7986 J。这个检查值不是第二小问所问的EFM。

HW7.3 闭合循环：先用恒流段连接状态

完整原题截图

HW7.3. SP 7.2

The flux linkage for a given electromechanical system is given as $\lambda = \frac{Ai}{x - x_0}$ where $A = 0.096$ Hm and $x_0 = 0.007$ m

$$\begin{bmatrix} & a & b & c & d & a \\ i & 0 & i_b & i_c & 0 & 0 \\ \lambda & 0 & 9 & \lambda_c & 0 & 0 \\ x & 0.047 & 0.047 & 0.014 & 0.014 & 0.047 \end{bmatrix}$$

The system is operated in a closed cycle $a \to b \to c \to d \to a$. x is held constant along $a \to b$ and $c \to d$ and i is held constant over $b \to c$ and $d \to a$.

What is i_c and λ_c?

i_c[A]=	number ±0.5%
λ_c[Wb turns]=	number ±0.5%

What is energy stored in the coupling field and the force of electric origin at point b?

W_{m_b}[J]=	number ±0.5%
f_b^e[N]=	number ±0.5%

来源：PrairieLearn HW7.3 实际网页题面。2026-11-02 截图。

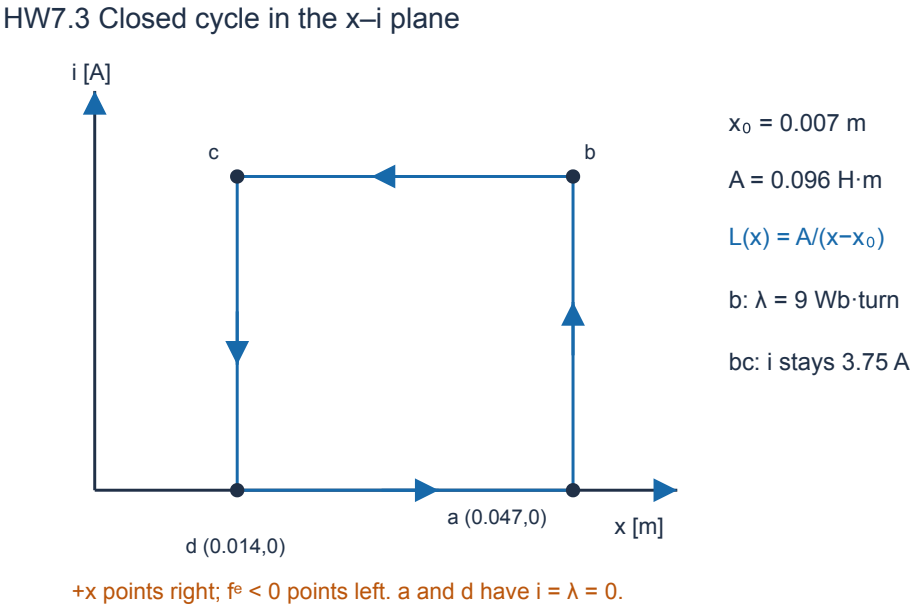
HW7.3 闭合循环：先用恒流段连接状态 · 解答

完整中文题意

线性本构λ=Ai/(x−x_0)，A=0.096 H·m、x_0=0.007 m。循环a→b→c→d→a的状态表为：a(i,λ,x)=(0,0,0.047)，b=(ib,9,0.047)，c=(ic,λc,0.014)，d=(0,0,0.014)，最后回a；电流单位A，磁链Wb·turn，位置m。a→b、c→d固定x，b→c、d→a固定i。求(a)ic与λc；(b)b点场能Wm,b及电磁力f^eb。

作答图与逐步求解

本题作答要画下列模型图，并保留图中方向、坐标、极性和参数。 原题截图不能代替作答图。



(a) 先解b，再用恒流关系到c：

$$
\begin{aligned}
i_b &= \frac{\lambda_b(x_b - x_0)}{A} = \frac{9(0.047 - 0.007)}{0.096} = 3.75\ \text{A},\\
i_c &= i_b = 3.75\ \text{A},\\
\lambda_c &= \frac{Ai_c}{x_c - x_0} = \frac{(0.096)(3.75)}{0.014 - 0.007} \approx 51.4286\ \text{Wb}\cdot\text{turn}.
\end{aligned}
$$

(b) 用b点，不能混入c点的较短间隙：

$$
\begin{aligned}
W_{m_b} &= \frac{1}{2}\lambda_b i_b = \frac{1}{2}(9)(3.75) = 16.875\ \text{J},\\
f_b^e &= \frac{1}{2}i_b^2 L'(x_b) = -\frac{Ai_b^2}{2(x_b - x_0)^2}\\
&= -\frac{(0.096)(3.75)^2}{2(0.047 - 0.007)^2} = -421.875\ \text{N}.
\end{aligned}
$$

负号沿−x。检查：b到c的有效间隙从0.040降到0.007 m，电流不变，所以λ按0.040/0.007放大；9×0.040/0.007=51.4286…。循环回原点只保证ΔWm=0，不保证EFE或EFM分别为零。

HW7.4 先固定磁链求力，再沿恒流路径积分

完整原题截图

HW7.4. 4.12

Given $W_m = 3x\lambda^2$, find the energy transfered from the mechanical system to the coupling field as x moves from 0.2 to 0.56 m along a constant current $i = 7$ A path.

EFM [J]= number ±0.5%

来源：PrairieLearn HW7.4 实际网页题面。2026–11–02 截图。

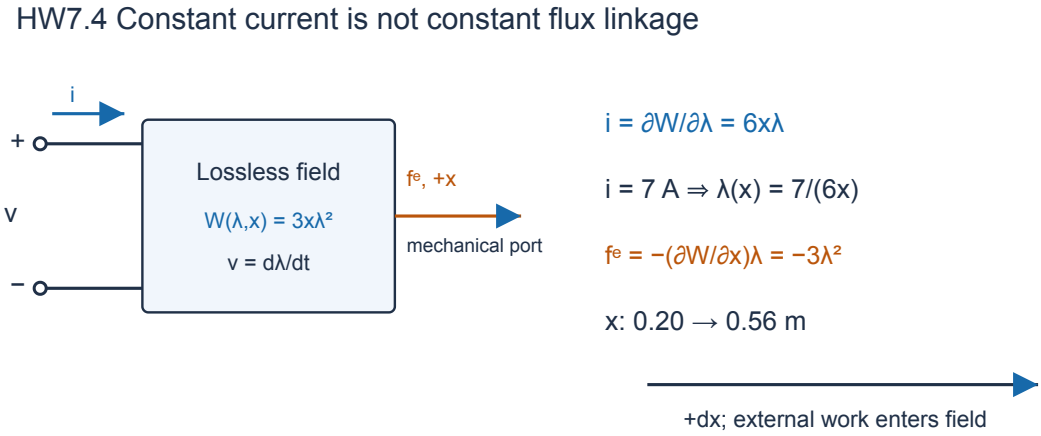
HW7.4 先固定磁链求力，再沿恒流路径积分 · 解答

完整中文题意

已知磁场能量Wm(λ,x)=3xλ²。电流沿全过程恒定为i=7 A，位置x由0.2 m移动到0.56 m。求从机械系统传入耦合场的能量EFM，单位J。

作答图与逐步求解

本题作答要画下列模型图，并保留图中方向、坐标、极性和参数。 原题截图不能代替作答图。



先以正确自然变量求导：

$$i=\left(\frac{\partial W_m}{\partial \lambda}\right)_x=6x\lambda, \qquad f^e=-\left(\frac{\partial W_m}{\partial x}\right)_\lambda=-3\lambda^2.$$

再使用恒流约束λ=7/(6x)：

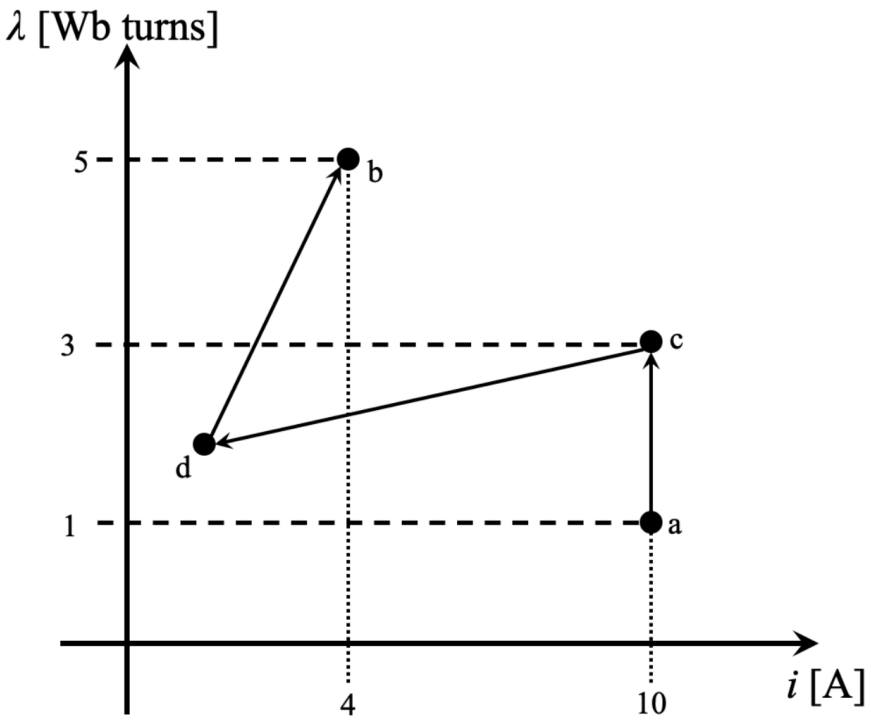
$$f^e(x)=-3\left(\frac{7}{6x}\right)^2=-\frac{49}{12x^2},$$
$$\mathrm{EFM}=-\int_{0.2}^{0.56} f^e(x)dx=\frac{49}{12}\int_{0.2}^{0.56} x^{-2}dx$$
$$=\frac{49}{12}\left[-\frac{1}{x}\right]_{0.2}^{0.56}=\frac{49}{12}\left(\frac{1}{0.2}-\frac{1}{0.56}\right)=+13.125\ \mathrm{J}.$$

前面的负号与负力相消，所以机械输入为正。检查：Wm=i²/(12x)，ΔWm=−13.125 J；EFE=iΔλ=−26.25 J；−26.25+13.125=−13.125 J。机械输入为正并不要求场能一定增加，因为电端同时输出更多能量。

HW7.5 λ−i折线：有符号梯形积分

完整原题截图

HW7.5. Energy Transfer EFE



A system with $\lambda = L(x)i$ is moved between points a and b through points c and d along straight line trajectories as shown . Point d is located at $\lambda = 2.8$ Wb turns and $i = 1.1$ A.

What is energy stored in the coupling field at points a and b?

W_{m_a} [J]=	number ±0.5%	?
W_{m_b} [J]=	number ±0.5%	?

What is the energy transfered from the electrical system to the coupling field when moving from a to b?

EFE [J]=	number ±0.5%	?

来源：PrairieLearn HW7.5 实际网页题面。2026-11-02 截图。

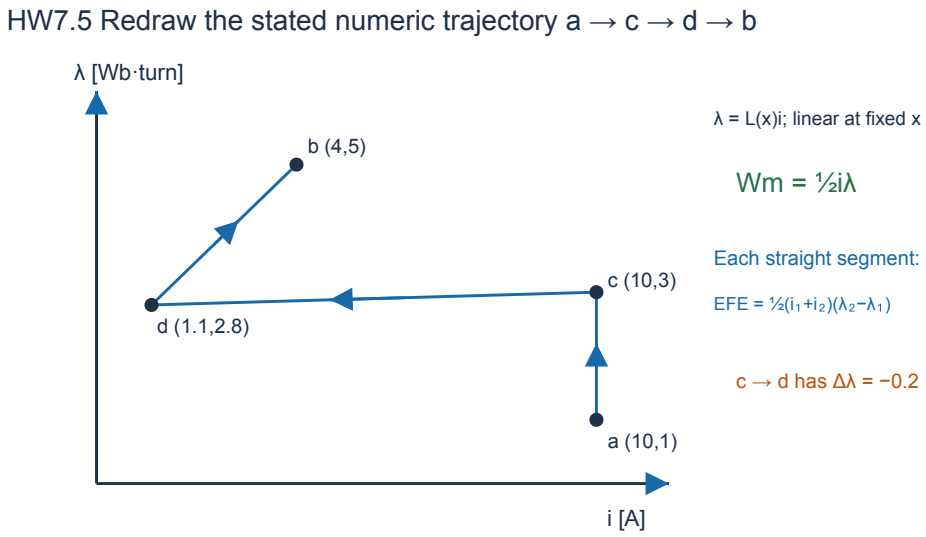
HW7.5 λ−i折线：有符号梯形积分 · 解答

完整中文题意

线性位置相关系统满足λ=L(x)i，按图沿a→c→d→b的直线段移动。图中a=(i,λ)=(10 A,1 Wb·turn)、c=(10 A,3 Wb·turn)、b=(4 A,5 Wb·turn)；文字另给d=(1.1 A,2.8 Wb·turn)。求(a)a、b点储能；(b)从a到b沿这条指定折线的电端输入EFE，单位J。原图d的示意位置不按这组文字参数的比例，计算以文字值为准。

作答图与逐步求解

本题作答要画下列模型图，并保留图中方向、坐标、极性和参数。 原题截图不能代替作答图。



Use d = (1.1 A, 2.8 Wb·turn) from the text; the source sketch is schematic.

(a) 两个端点各用自己的i、λ：

$$W_{m_a} = \frac{1}{2} i_a \lambda_a = \frac{1}{2}(10)(1) = 5\ \text{J}, \qquad W_{m_b} = \frac{1}{2} i_b \lambda_b = \frac{1}{2}(4)(5) = 10\ \text{J}.$$

(b) 分三段，不能直接把a、b连成一段：

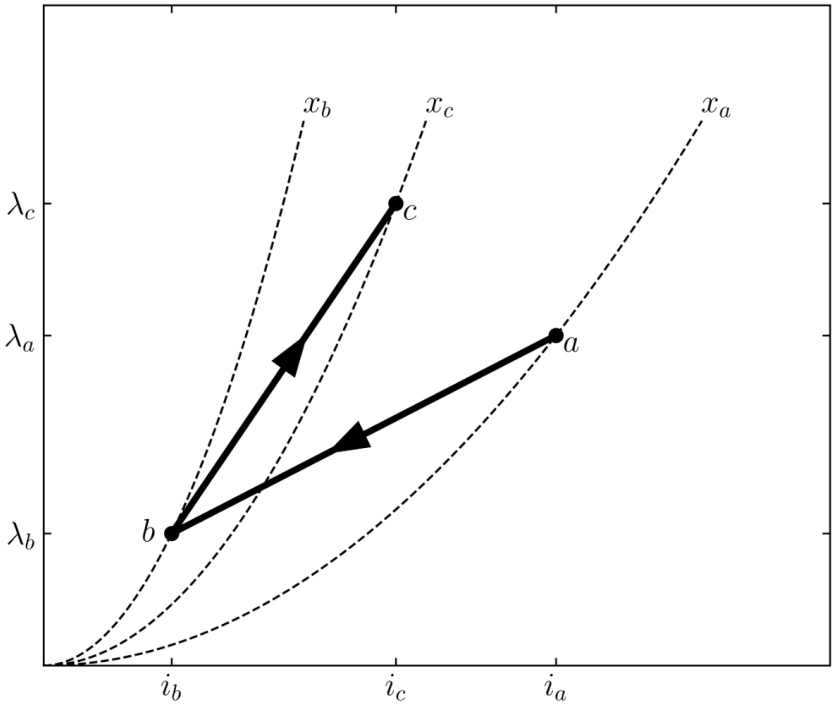
$$\begin{aligned}
\text{EFE}_{a\to c} &= \frac{i_a + i_c}{2}(\lambda_c - \lambda_a) = \frac{10+10}{2}(3-1) = 20\ \text{J},\\
\text{EFE}_{c\to d} &= \frac{i_c + i_d}{2}(\lambda_d - \lambda_c) = \frac{10+1.1}{2}(2.8-3) = -1.11\ \text{J},\\
\text{EFE}_{d\to b} &= \frac{i_d + i_b}{2}(\lambda_b - \lambda_d) = \frac{1.1+4}{2}(5-2.8) = 5.61\ \text{J},\\
\text{EFE}_{a\to b} &= 20 - 1.11 + 5.61 = +24.5\ \text{J}.
\end{aligned}$$

c→d的磁链减少，故这一段电端能量为负；总量仍为正。检查：ΔWm=5 J，机械输入应为5−24.5=−19.5 J，即有19.5 J净机械输出；不能把EFE误答成ΔWm。

HW7.6 二次电流本构：两端状态的磁能

完整原题截图

HW7.6. Energy Transfer EFM (part 1)



A system with $\lambda = \frac{A}{x - x_0} i^2$ where $A = 0.005$ Wb t m/A^2 and $x_0 = 0.06$ m moves from point a to point c through point b with the points defined as:

Point a: $\lambda_a = 5.43$ Wb t, $i_a = 7.8$ A, $x_a = 0.116$ m

Point b: $\lambda_b = 1.67$ Wb t, $i_b = 1$ A, $x_b = 0.063$ m

Point c: $\lambda_c = 8.86$ Wb t, $i_c = 4.8$ A, $x_c = 0.073$ m

What is energy stored in the coupling field at points a and c?

W_{m_a}[J]=	number ±0.5%	?
W_{m_c}[J]=	number ±0.5%	?

来源：PrairieLearn HW7.6 实际网页题面。2026-11-02 截图。

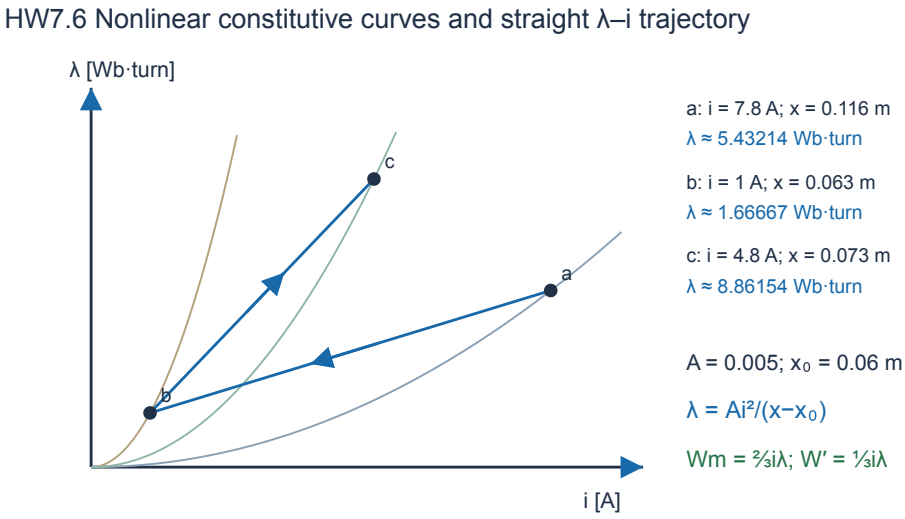
HW7.6 二次电流本构：两端状态的磁能 · 解答

完整中文题意

系统满足λ=A i²/(x−x₀)，A=0.005 Wb·turn·m/A²、x₀=0.06 m，沿a→b→c运动。题给a：λa=5.43 Wb·turn、ia=7.8 A、xa=0.116 m；b：λb=1.67、ib=1 A、xb=0.063 m；c：λc=8.86、ic=4.8 A、xc=0.073 m。求a、c点耦合磁场储能，单位J。λ行是两位小数显示值，公式和i、x能重建更精确磁链。

作答图与逐步求解

本题作答要画下列模型图，并保留图中方向、坐标、极性和参数。 原题截图不能代替作答图。



Thin curves: fixed x. Thick segments: a → b → c, straight in the i–λ plane.

Positive EFE and EFM point into the field; ΔWm = EFE + EFM.

$$W'_m = \int_0^i \frac{A\hat{i}^2}{x - x_0} d\hat{i} = \frac{Ai^3}{3(x - x_0)}, \qquad W_m = i\lambda - W'_m = \frac{2Ai^3}{3(x - x_0)}.$$

按本构重建端点：

$$\lambda_a = \frac{Ai_a^2}{x_a - x_0} = \frac{(0.005)(7.8)^2}{0.116 - 0.06} \approx 5.43214285714 \text{ Wb} \cdot \text{turn},$$

$$\lambda_c = \frac{Ai_c^2}{x_c - x_0} = \frac{(0.005)(4.8)^2}{0.073 - 0.06} \approx 8.86153846154 \text{ Wb} \cdot \text{turn},$$

$$W_{m_a} = \frac{2Ai_a^3}{3(x_a - x_0)} = \frac{2(0.005)(7.8)^3}{3(0.116 - 0.06)} \approx 28.2471 \text{ J},$$

$$W_{m_c} = \frac{2Ai_c^3}{3(x_c - x_0)} = \frac{2(0.005)(4.8)^3}{3(0.073 - 0.06)} \approx 28.3569 \text{ J}.$$

▼ 详情

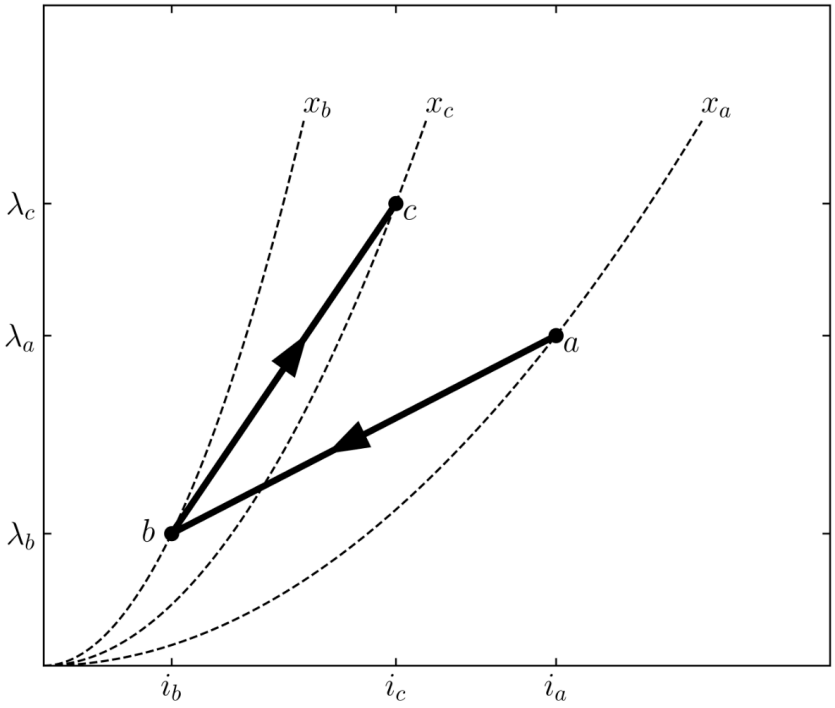
$W_{m_a} \approx \frac{2}{3}(7.8)(5.43) = 28.236$ J；$W_{m_c} \approx \frac{2}{3}(4.8)(8.86) = 28.352$ J。差异来自题面λ只显示两位小数；主结果使用未提前舍入的本构计算。原题数值全部保留，未声称平台采用哪组评分常数。

检查：Wm/W′m=2，Wm+W′m=iλ；与HW6.5的三次磁链不同，本题系数不是¾。

HW7.7 非线性折线：由能量守恒求机械输入

完整原题截图

HW7.7. Energy Transfer EFM (part 2)



A system with $\lambda = \frac{A}{x - x_0} i^2$ where $A = 0.003$ Wb t m/A^2 and $x_0 = 0.04$ m moves from point a to point c through point b with the points defined as:

Point a: $\lambda_a = 4.23$ Wb t, $i_a = 7.6$ A, $x_a = 0.081$ m

Point b: $\lambda_b = 2.61$ Wb t, $i_b = 2.8$ A, $x_b = 0.049$ m

Point c: $\lambda_c = 7.93$ Wb t, $i_c = 4.6$ A, $x_c = 0.048$ m

The energy stored in the coupling field at points a and c are given below.

$W_m(a) = 21.413463414634144$ J

$W_m(c) = 24.333999999999996$ J

The system moves from point a to point b along a straight line and from point b to point c along a straight line.

What is the energy transfered from the mechanical system to the coupling field when moving from a to c?

EFM [J]= number ±0.5%

来源：PrairieLearn HW7.7 实际网页题面。2026–11–02 截图。

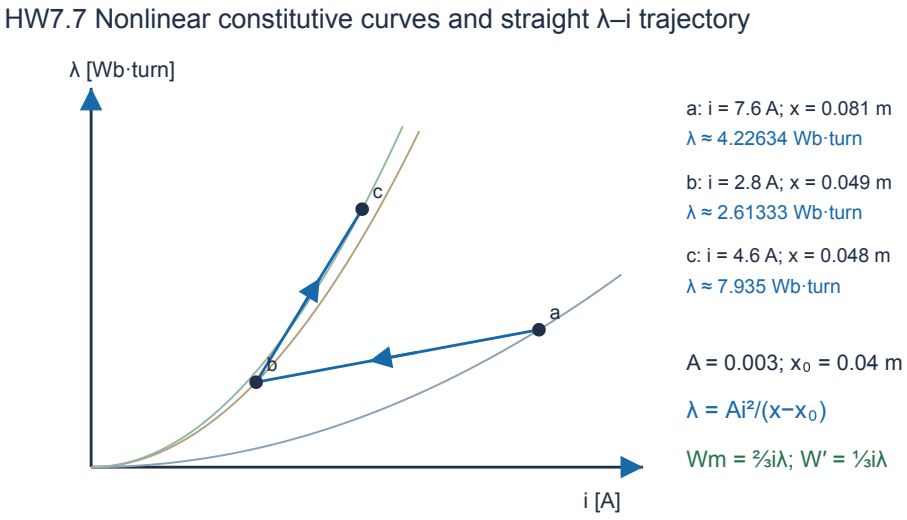
HW7.7 非线性折线：由能量守恒求机械输入 · 解答

完整中文题意

系统满足λ=A i²/(x−x₀)，A=0.003 Wb·turn·m/A²、x₀=0.04 m。a：λa=4.23 Wb·turn、ia=7.6 A、xa=0.081 m；b：λb=2.61、ib=2.8 A、xb=0.049 m；c：λc=7.93、ic=4.6 A、xc=0.048 m。题目另给Wm(a)=21.413463414634144 J、Wm(c)=24.333999999999996 J。系统在λ−i图上沿直线段a→b、b→c运动，求从机械系统传入耦合场的净能量EFM，单位J。本题参数独立于HW7.6，不能直接借用上一题的数值。

作答图与逐步求解

本题作答要画下列模型图，并保留图中方向、坐标、极性和参数。 原题截图不能代替作答图。



先保留题给场能：

$$\Delta W_m = W_m(c) - W_m(a) = 24.333999999999996 - 21.413463414634144 \approx 2.92053658537\ \text{J}.$$

由本构得到较完整的磁链，不提前取两位小数：

$$\lambda_a = \frac{Ai_a^2}{x_a - x_0} = \frac{(0.003)(7.6)^2}{0.081 - 0.04} \approx 4.22634146341,$$

$$\lambda_b = \frac{Ai_b^2}{x_b - x_0} = \frac{(0.003)(2.8)^2}{0.049 - 0.04} \approx 2.61333333333,$$

$$\lambda_c = \frac{Ai_c^2}{x_c - x_0} = \frac{(0.003)(4.6)^2}{0.048 - 0.04} = 7.935 \quad (\text{Wb} \cdot \text{turn}).$$

$$\mathrm{EFE}_{a\to b} = \frac{i_a + i_b}{2}(\lambda_b - \lambda_a) \approx \frac{7.6+2.8}{2}(2.61333333333 - 4.22634146341) \approx -8.38764227642\ \mathrm{J},$$
$$\mathrm{EFE}_{b\to c} = \frac{i_b + i_c}{2}(\lambda_c - \lambda_b) \approx \frac{2.8+4.6}{2}(7.935 - 2.61333333333) \approx 19.6901666667\ \mathrm{J},$$
$$\mathrm{EFE}_{a\to c} = \mathrm{EFE}_{a\to b} + \mathrm{EFE}_{b\to c} \approx 11.3025243902\ \mathrm{J},$$
$$\mathrm{EFM} = \Delta W_m - \mathrm{EFE} \approx 2.92053658537 - 11.3025243902 \approx -8.38199\ \mathrm{J}.$$

净机械输入为负，即磁场向机械系统净输出8.38199 J；不能把它改报正的EFM。

▼ 详情

若将显示λ按精确端点使用，$\mathrm{EFE} = \frac{7.6+2.8}{2}(2.61 - 4.23) + \frac{2.8+4.6}{2}(7.93 - 2.61) = 11.26$ J；结合题给场能得到 $\mathrm{EFM} \approx -8.33946$ J。这组显示λ与本构/给定场能不能同时严格成立。主结果−8.38199 J使用本构精度，与已给场能一致；备用结果保留“按网页显示值字面计算”的可审查依据，不擅改原题或推断平台判分规则。

独立检查：沿每段以t参数化i(t)、λ(t)，由$x = x_0 + Ai^2/\lambda$求dx/dt，直接数值积分$-\int f^e dx$，其中 $f^e = -Ai^3/[3(x - x_0)^2]$，与上述守恒结果一致。